

Table 25: Answer table for Q.28 over Table 2 of Kim et al. [55], modeled in the ORKG as Comparison R1563131. The table lists metal ALE processes with $EPC \geq 2$ /cycle, including direction, modification and removal chemistries, process temperature, and cycle time.

Material	Direction	Reaction	Removal	EPC	Temp ($^{\circ}$ C)	Cycle time (sec)
Fe	Anisotropy	O ₂	Formic acid	42	80	350
Co	Anisotropy	O ₂	Formic acid	28	80	350
Pd	Anisotropy	O ₂	Formic acid	12	80	350
W	Isotropy	O ₂	WF ₆	6.3	300	320
Pt	Anisotropy	O ₂	Formic acid	5	80	350
W	Anisotropy	NF ₃	O ₂	2.6		40
W	Isotropy	O ₂ /O ₃	BCl ₃ →HF	2.5	207	325
W	Anisotropy	Cl ₂	Ar	2.1	60	25
Co	Isotropy	Cl ₂	hfacH	2	140	95

Easy Query. The 28th question is: “List all metal ALE processes with $EPC \geq 2$ /cycle; return the material, direction (anisotropic vs. isotropic), modification and removal chemistries, EPC, process temperature, and cycle time; sort by EPC in descending order.” The SPARQL query implementation for this question can be rerun via <https://tinyurl.com/t2-metals-high-epc>. The resulting high-EPC metal ALE processes are shown in Table 25.

Table 26: Answer table for Q.29 over Table 2 of Kim et al. [55], modeled in the ORKG as Comparison R1563131. For each etching direction, the table reports the number of distinct metals and the mean EPC (in $\text{\AA}$ /cycle).

Direction	Number of Metals	Mean EPC ($\text{\AA}$ /cycle)
Anisotropy	7	11.6875
Isotropy	5	2.035

Complex Query. The 29th question is: “Group metal ALE processes by direction (anisotropic vs. isotropic) and compute, for each group, the number of distinct metals covered and the mean EPC; sort by mean EPC in descending order.” The SPARQL query implementation for this question can be rerun via <https://tinyurl.com/t2-metals-complex>. The aggregated results are shown in Table 26.

3.9 Paper – Atomic layer etching of SiO₂ for nanoscale semiconductor devices: A review

Hong *et al.* (2024) [27] review ALE strategies for SiO₂ in advanced semiconductor patterning. They contrast fluorocarbon-based anisotropic ALE—controlled by precursor choice, ion energy, etch selectivity, and chamber-wall effects—with isotropic ALE routes that convert SiO₂ into volatile intermediates such as Al₂O₃ or ammonium fluorosilicate under thermal or plasma activation. By linking mechanistic steps to device-relevant process metrics, the review discusses how SiO₂ ALE can be tuned for emerging 3D architectures and sub-10 nm fabrication requirements.

3.9.1 Machine-actionable modeling of Tables I–VI

Tables I–VI in the SiO₂ ALE review together cover fluorocarbon-based anisotropic ALE and thermal or plasma-assisted isotropic mechanisms. Tables I and II (<https://orkg.org/comparisons/R1560949> [28], <https://orkg.org/comparisons/R1560977> [29]) list fluorination precursors, removal gases, etch rates, temperatures, and ion-energy windows for anisotropic SiO₂ ALE. Tables III and IV (<https://orkg.org/comparisons/R1561025> [30], <https://orkg.org/comparisons/R1561023> [31]) summarize selectivity (SiO₂/Si and SiO₂/Si₃N₄), methods for improving selectivity, and chamber-wall conditioning effects. Tables V and VI (<https://orkg.org/comparisons/R1561046> [32], <https://orkg.org/comparisons/R1562478> [33]) cover thermal and plasma-assisted isotropic ALE, organizing multi-step reaction pathways together with etch rates.

As in the other ALE tables modeled in this work, all six SiO₂ tables are represented in the ORKG as machine-actionable comparisons using a shared, harmonized predicate schema. For Tables I–

IV, core descriptors include P183129 (fluorination precursor chemistry), P183125 (removal chemistry), P183130 (process temperature), and P183131 (etch rate), with Table II adding the plasma-relevant P183132 (ion energy in the removal step). Table III introduces selectivity-focused predicates P183133 (selectivity material pair), P173032 (selectivity value), and P183134 (selectivity improvement method), while Table IV contributes P183135 (method of chamber-wall effect). The isotropic ALE mechanism tables (V and VI) use P183136 (etching mechanism class) together with step-level predicates P183137, P183138, and P183139. Within this shared schema, cross-table SPARQL queries can link precursor families, ion-energy windows, selectivity ranges, chamber-wall treatments, and isotropic mechanism pathways across the entire ALE corpus.

The six SiO₂ ALE comparisons are also curated in an ORKG Smart Review (<https://orkg.org/reviews/R1562498>), which provides a single entry point to the modeled tables from Hong *et al.* (2024) [27]. While cross-table SPARQL queries can be executed directly on the comparisons, the Smart Review documents the modeling choices, highlights example queries, and connects the semantic data model back to the original article.

Table 27: Answer table for Q.30 (Easy Cross-Query 1) over Tables I and II of Hong *et al.* [27], modeled in the ORKG as Comparisons R1560949 and R1560977. The table lists fluorocarbon ALE processes that appear in both tables under near-room-temperature (RT $\pm$ 20°C) conditions, together with removal chemistry, etching rate, and ion-energy window.

Precursor	Removal	Temp (°C)	EPC (Å/cycle)	Ion-energy window
C4F6/Ar plasma	Ar plasma	−10	14.2	10–100 V
CHF ₃ plasma	Ar plasma	−40–20	9.0	0–50 V
C4F8 plasma	Ar plasma	20	5.8	30–90 V
CF ₃ CF ₂ CF ₂ CH ₂ OH plasma	Ar plasma	RT	5.2	10–80 V
CHF ₃ plasma	O ₂ plasma	20	4.1	30–90 V
n-C ₃ F ₇ OCH ₃ plasma	Ar plasma	RT	2.1	10–80 V
C4F8 plasma	Ar ion beam	RT	1.9	30–200 V
i-C ₃ F ₇ OCH ₃ plasma	Ar plasma	RT	1.8	10–80 V

Easy Cross-Query 1. The 30th question is: “For each fluorocarbon precursor system that appears in both Table I and Table II, list the precursor chemistry, removal gas, process temperature, etching rate, and ion-energy window used in the removal step; restrict to conditions near room temperature (RT $\pm$ 20°C) and sort by etching rate in descending order.” The SPARQL query implementation for this question can be rerun via <https://tinyurl.com/pap6-crossq1-easy>. The resulting fluorocarbon ALE processes are shown in Table 27.

Table 28: Answer table for Q.31 (Easy Cross-Query 2) over Tables III and IV of Hong *et al.* [27], modeled in the ORKG as Comparisons R1561025 and R1561023. The table lists C₄F₈/Ar-based anisotropic SiO₂ ALE processes that appear in both tables, including their selectivity pair, selectivity range, selectivity method, chamber-wall treatment, and etch rate.

Selectivity pair	Selectivity	Selectivity method	Chamber-wall treatment	EPC (Å/cycle)
SiO ₂ /Si ₃ N ₄	0.2–15.0	Precursor selection; Fluorocarbon film thickness; Ion energy; Etching step time	Chamber cleaning with O ₂ plasma; Chamber wall heating	3
SiO ₂ /Si ₃ N ₄	> 7.0	Precursor selection	Chamber cleaning with O ₂ plasma; Chamber wall heating	3
SiO ₂ /Si	> 10.0	Precursor selection	Chamber cleaning with O ₂ plasma; Chamber wall heating	3

Easy Cross-Query 2. The 31st question is: “Collect all anisotropic SiO₂ ALE processes based on C₄F₈/Ar plasma from Table III and Table IV, and list for each the target selectivity pair (SiO₂/Si or SiO₂/Si₃N₄), selectivity range (if available), chamber-wall treatment (none, O₂ cleaning, wall heating), and etching rate; sort by etching rate.” The SPARQL query implementation for this question can be rerun via <https://tinyurl.com/pap6-crossq2-easy>. The matched processes are shown in Table 28.

Table 29: Answer table for Q.32 (Complex Cross-Query 1) over Tables I–III of Hong et al. [27], modeled in the ORKG as Comparisons R1560949, R1560977, and R1561025. For each fluorocarbon precursor family, the table reports mean and maximum SiO₂ etch rate, the combined ion-energy windows, and the maximum reported SiO₂/Si and SiO₂/Si₃N₄ selectivity.

Precursor family	Mean etch rate	Max. etch rate	Ion-energy windows	Max. Sel. SiO ₂ /Si	Max. Sel. SiO ₂ /Si ₃ N ₄
C ₄ F ₈	—	—	—	0	0
C ₃ H ₃ F ₃	—	—	—	0	0
C ₄ F ₆	—	—	10–100 V	—	—
CF ₃ I	—	—	—	—	—
CHF ₃	—	—	—	0	0

Complex Cross-Query 1. The 32nd question is: “Group all anisotropic SiO₂ ALE processes by fluorocarbon precursor family (e.g., C₄F₈-based, CHF₃-based, C₃F₇OCH₃ isomers) across Tables I, II, and III. For each family, compute (i) the mean and maximum SiO₂ etching rate, (ii) the union of reported ALE ion-energy windows in the removal step, and (iii) the maximum reported SiO₂/Si and SiO₂/Si₃N₄ selectivity; return one row per precursor family, ordered first by maximum selectivity, then by mean etching rate.” The SPARQL query implementation for this question can be rerun via <https://tinyurl.com/pap6-crossq1-complex>. The resulting precursor-family aggregation is shown in Table 29, and reflects the limited amount of parseable numeric etch-rate and selectivity data currently available in the modeled tables.

Table 30: Answer table for Q.33 (Complex Cross-Query 2) over Tables V and VI of Hong et al. [27], modeled in the ORKG as Comparisons R1561046 and R1562478. Isotropic SiO₂ ALE processes are grouped by etching mechanism class; for each class, the table reports the number of variants, minimum/maximum/mean etching rate, temperature range associated with rate-limiting steps, and whether plasma is required and high-rate operation at low temperature is achieved.

Mechanism class	# variants	Min. rate	Max. rate	Mean rate	Temp range [°C]	Plasma required	High-rate, low- <i>T</i>
AFS-based plasma-assisted	3	27	75	43	3–4	yes	yes
AFS-based thermal	1	9	9	9	20–20	no	yes
Al ₂ O ₃ -conversion cycles	2	0.07	0.35	0.21	300–350	no	no

Complex Cross-Query 2 (Tables V + VI: thermal vs. plasma-assisted isotropic ALE). The 33rd question is: “Combine all isotropic SiO₂ ALE processes and group by mechanism class; report number of variants, min/max/mean etching rate, temperature range, and plasma requirement.” The SPARQL query implementation for this question can be rerun via <https://tinyurl.com/pap6-crossq2-complex-fixed>. The aggregated mechanism-class statistics are shown in Table 30.

With this, we conclude the section on ALE machine-actionable comparisons in the ORKG.

Across all ALD and ALE review papers considered in this work, we modeled a total of **18 machine-actionable tables** as ORKG Comparisons: **seven** from three ALD papers and **eleven** from six ALE papers. These structured representations provide a unified semantic layer for precursor and reactant chemistries, process parameters, etch and growth metrics, selectivity behavior, ion-energy windows, chamber-wall effects, and isotropic multi-step mechanisms. In total, **33 natural-language questions**—**19** for ALD and **14** for ALE—were implemented as SPARQL queries, each producing a reproducible, machine-derived result table. Together, these queries form a comprehensive benchmark for neurosymbolic querying over materials-science knowledge in the ORKG.

As supplementary data to this paper, a consolidated catalog of papers, tables, comparisons, NL questions, and SPARQL links is available in our GitHub repository: <https://github.com/sciknoworg/ald-ale-orkg-review>.